

ALLIANCE UNIVERSITY

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NAAC GRADE A+ ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No: 6

Title: Sorting & Filtering Data

GITHUB LINK

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1. Problem Statement

In modern enterprise sales operations, raw transaction logs capture thousands of data points encompassing diverse product lines, geographic territories, customer types, and discounting structures. Without systematic data organization, sales leadership faces significant challenges in distinguishing lucrative growth drivers from silent profit drains. In this scenario, a sales manager seeks to extract critical operational business intelligence by isolating the company's highest-value transactions and diagnosing underperforming products. To accomplish this, advanced sorting and filtering techniques—including multi-level sorting, numerical boundary filtering, conditional status evaluation, and dynamic lookup modeling—must be systematically applied to transform transactional data into executive decision support.

2. Objectives

- Apply multi-level sorting techniques to rank transactions by revenue and profitability.
- Implement numerical boundary filtering to isolate high-value deals meeting the 90th percentile threshold (Revenue >= $32,400).
- Formulate rigorous dual-threshold criteria (Revenue < 60% of product average OR Profit Margin < 15%) to filter and flag underperforming products.
- Deliver actionable executive recommendations to safeguard margins, optimize pricing, and accelerate enterprise revenue growth.

3. Dataset Architecture & Description

The transactional dataset encompasses 300 business-to-business (B2B) sales transactions executed across Fiscal Year 2026 (October 2025 through September 2026). The operational environment spans four geographic regions (East, West, North, South), seven account executives, twelve corporate clients, and twelve distinct products categorized across Hardware, Software, and Professional Services.

Overall Dataset Summary: Across all 300 transactions, total gross revenue reached $4,289,922.00, generating $1,706,242.00 in gross profit with an overall company profit margin of 39.77%. A total of 2,264 units were sold, with an average transaction value of $14,299.74 and average product revenue of $357,493.50.

4. Methodology: Sorting & Filtering Techniques Applied

The investigation was conducted utilizing systematic data analytics procedures across spreadsheet modeling and data transformation pipelines:

4.1 Multi-Level & Custom Sorting

Sorting rearranges tabular records based on specified criteria without altering the underlying data. In this analysis, multi-tiered sorting was executed:
1. Primary Sort by Revenue (Descending): Positions the highest revenue-generating transactions at the top, enabling immediate discovery of high-impact accounts.
2. Secondary Sort by Profit (Descending): Resolves ties and ranks deals by realized net gross profit.
3. Ascending Revenue Sort for Underperformers: Orders troubled product lines from lowest to highest revenue, spotlighting severe commercial deficits first.

4.2 Filtering Mechanisms & Numerical Boundary Rules

Filtering temporarily hides rows that do not satisfy explicit logical conditions, isolating subsets for targeted analysis:

• AutoFilter Application: Deployed across all column headers in the transaction ledger, allowing instantaneous dimensional slicing by Region, Product, Customer, and Return status.

• 90th Percentile Threshold Filter: Applied a numerical condition [Revenue >= $32,400] to extract the top decile of transactions representing enterprise-tier contracts.

• Quality & Return Filtering: Sliced by [Returned = 'Y'] to isolate reversed transactions and assess phantom revenue bookings.

5. Analysis of Highest-Value Transactions

Using sorting (Revenue Descending) and numerical filtering (Revenue >= $32,400), exactly 32 high-value transactions were extracted from the 300 total records.

5.1 Key Findings: Top-Decile Revenue Concentration

- Revenue Concentration: The 32 high-value transactions represent only 10.67% of total transaction volume, yet they contribute $1,187,172.50 in revenue (27.67% of total company revenue) and $477,572.50 in profit (27.99% of total profit).
- Peak Transaction: The single largest transaction is TXN-1156, closed by Priya Nair in the South region for Juniper Foods, purchasing 4 units of Enterprise CRM Suite at 0% discount, generating $48,000.00 in revenue and $28,000.00 in profit.
- Critical Return Audit Finding: Transaction TXN-1124 ($40,800.00 revenue, $20,800.00 profit) was flagged as Returned ('Y'). When filtering for realized net revenue, this record must be deducted, adjusting net high-value revenue to $1,146,372.50.

5.2 Top 15 Highest-Value Transactions

Table 1 displays the Top 15 transactions sorted strictly in descending order of revenue.

Rank	Txn ID	Date	Region	Salesperson	Customer	Product	Revenue ($)
1	TXN-1156	2026-04-05	South	Priya Nair	Juniper Foods	Enterprise CRM Suite	$48,000.00
2	TXN-1227	2026-06-18	North	Aisha Khan	BlueWave Ltd	Onboarding Package	$45,000.00
3	TXN-1119	2026-02-24	West	Sofia Rossi	Granite Bank	Enterprise CRM Suite	$43,200.00
4	TXN-1204	2026-05-22	North	Aisha Khan	Lumen Telecom	Enterprise CRM Suite	$43,200.00
5	TXN-1081	2026-01-07	East	Marcus Hill	Cobalt Systems	Onboarding Package	$42,750.00
6	TXN-1107	2026-02-06	North	Tom Becker	Cobalt Systems	Onboarding Package	$42,750.00
7	TXN-1124	2026-02-27	West	Daniel Cho	Lumen Telecom	Enterprise CRM Suite	$40,800.00

8	TXN-1019	2025-10-31	West	Sofia Rossi	Granite Bank	Server Rack X900	$39,200.00
9	TXN-1214	2026-05-31	East	Marcus Hill	Granite Bank	Server Rack X900	$39,200.00
10	TXN-1222	2026-06-13	North	Aisha Khan	Keystone Edu	Server Rack X900	$39,200.00
11	TXN-1229	2026-06-20	East	Marcus Hill	Ionic Energy	Enterprise CRM Suite	$38,400.00
12	TXN-1008	2025-10-15	East	Marcus Hill	Orion Travel	Business Laptop L5	$36,250.00
13	TXN-1200	2026-05-19	North	Tom Becker	Lumen Telecom	Business Laptop L5	$36,250.00
14	TXN-1086	2026-01-17	West	Daniel Cho	Fusion Media	Enterprise CRM Suite	$36,000.00
15	TXN-1091	2026-01-24	South	Luis Ortega	Juniper Foods	Onboarding Package	$36,000.00

6. Analysis of Underperforming Products

A comprehensive product evaluation was conducted across all 12 products. Underperformance was established via dual quantitative criteria:
1. Revenue Criterion: Total product revenue less than 60% of average product revenue (< $214,496.10, where average is $357,493.50).
2. Profitability Criterion: Profit margin less than 15.0% (or negative gross profit).

6.1 Full Product Portfolio Performance Summary

Table 2 summarizes the operational performance of all 12 product lines, sorted by gross revenue.

Product	Category	Txns	Units	Revenue ($)	Profit ($)	Margin %	Status
Business Laptop L5	Hardware	60	638	$849,192.50	$179,292.50	21.11%	OK
Analytics Platform	Software	29	76	$586,925.00	$343,725.00	58.56%	OK
Server Rack X900	Hardware	21	62	$570,360.00	$123,960.00	21.73%	OK
Onboarding Package	Services	24	318	$532,530.00	$214,530.00	40.29%	OK
Enterprise CRM Suite	Software	14	43	$465,000.00	$250,000.00	53.76%	OK
Cybersecurity Shield	Software	25	69	$387,500.00	$208,100.00	53.70%	OK
Cloud Backup Pro	Software	37	162	$355,560.00	$209,760.00	58.99%	OK
Premium Support Plan	Services	16	83	$268,275.00	$152,075.00	56.69%	OK
Wireless Router W2	Hardware	61	706	$206,512.00	$37,072.00	17.95%	Underperforming
Smart Display 55in	Hardware	5	24	$33,155.00	$-6,445.00	-19.44%	Underperforming
Basic Training Kit	Services	5	61	$26,970.00	$-4,750.00	-17.61%	Underperforming
Legacy Fax Gateway	Hardware	3	22	$7,942.50	$-1,077.50	-13.57%	Underperforming

6.2 Diagnosing the Underperforming Products

Filtering the product portfolio revealed four underperforming products. Three are actively loss-making, while the fourth suffers from severe revenue dilution. Discounts of 20% to 30% on products with thin baseline margins (about 9% to 13%) pushed realized prices below unit cost.

Product	Category	Units	Revenue ($)	Profit ($)	Margin %	Primary Deficit
Legacy Fax Gateway	Hardware	22	$7,942.50	$-1,077.50	-13.57%	Negative Margin (Loss)
Basic Training Kit	Services	61	$26,970.00	$-4,750.00	-17.61%	Negative Margin (Loss)
Smart Display 55in	Hardware	24	$33,155.00	$-6,445.00	-19.44%	Negative Margin (Loss)
Wireless Router W2	Hardware	706	$206,512.00	$37,072.00	17.95%	Low Ticket Size

7. Strategic Recommendations for Sales Leadership

1. Enforce Automated Minimum Margin Discount Guardrails: Eliminate discretionary rep discounting. Set hard discount ceilings: Smart Display 55in max discount 5%, Basic Training Kit max discount 5%. Require VP approval for any discount that pushes gross margin below 20%.
1. Immediate Sunsetting of Legacy Fax Gateway: Retire the product immediately. With only 3 transactions, declining revenue, and negative margins, continuing to stock and support this legacy hardware is unviable.
1. Restructure Service Delivery Costs for Basic Training Kit: Shift training delivery to on-demand digital modules or raise list price to $850 to ensure gross margin exceeds 40% even after enterprise discounts.
1. Reposition Wireless Router W2 into Enterprise Bundles: Discontinue standalone single-unit router sales. Bundle routers exclusively with Business Laptops or Server Racks as a networking enablement package, raising average deal size.

8. Conclusion

Through the systematic application of sorting and filtering techniques in Microsoft Excel and data analytics workflows, the company's operational sales data was rigorously evaluated. Multi-level descending revenue sorting successfully isolated 32 enterprise-grade transactions accounting for 27.67% of revenue and 27.99% of profit. Concurrently, dual-threshold filtering diagnosed four underperforming products, uncovering $12,272.50 in direct losses stemming from uncontrolled discounting on low-margin goods. These findings provide sales management with clear, empirical evidence to institute discount governance, retire obsolete lines, and protect core commercial assets.